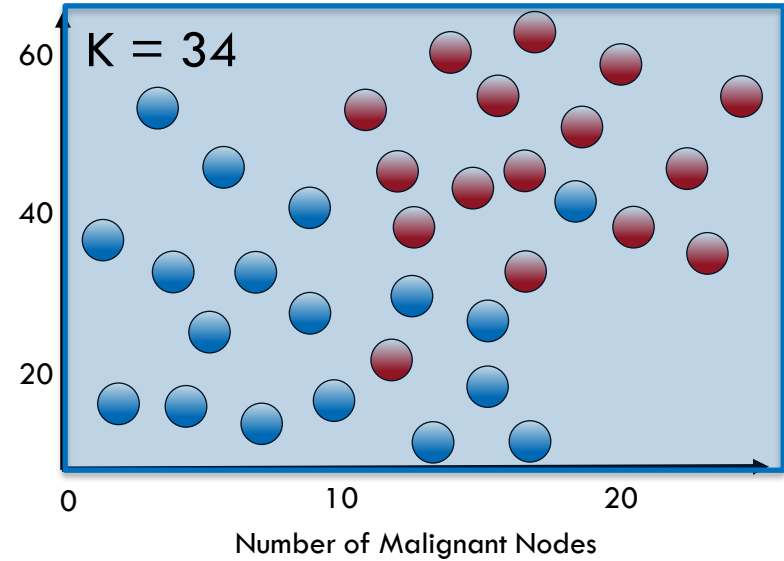
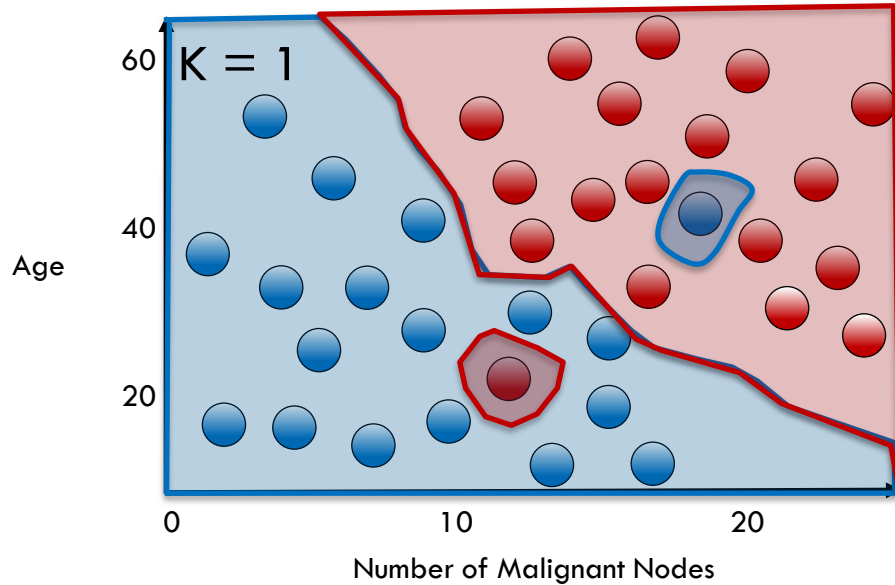

Model Generalization

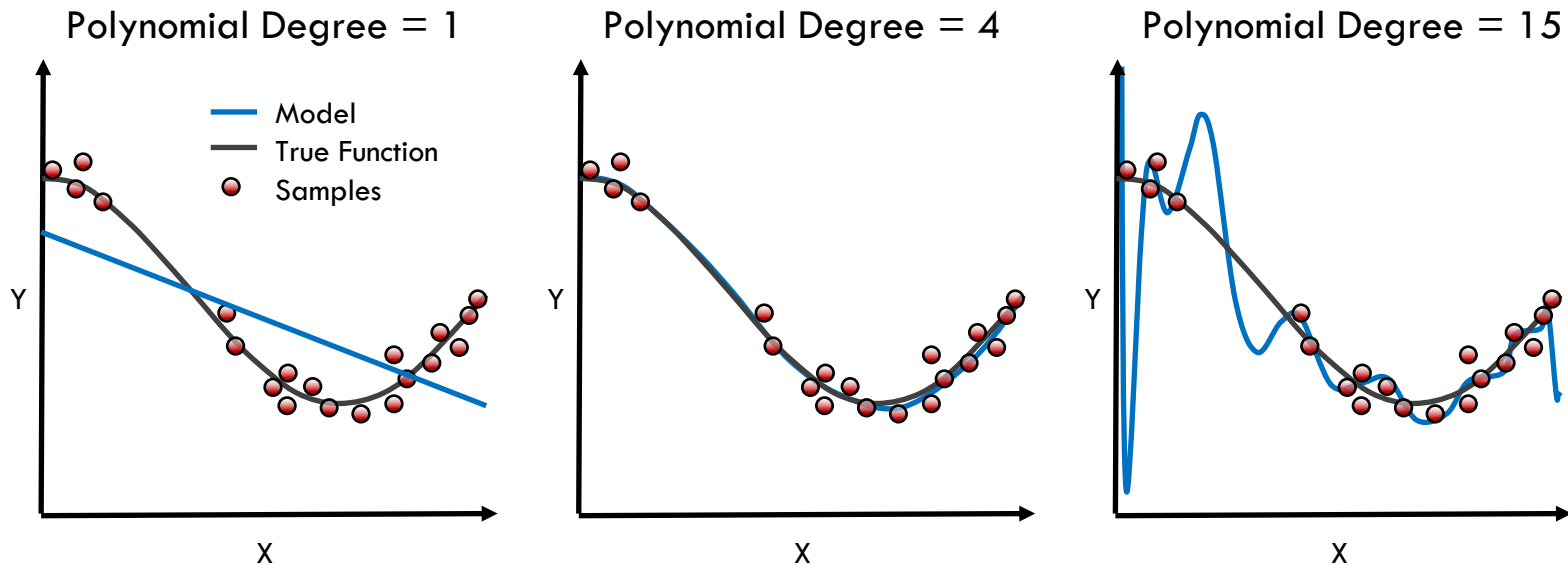
Learning Objectives

- Explain the difference between over-fitting and under-fitting a model
- Describe Bias-variance tradeoffs
- Find the optimal training and test data set splits, cross-validation, and model complexity versus error
- Apply a linear regression model for supervised learning

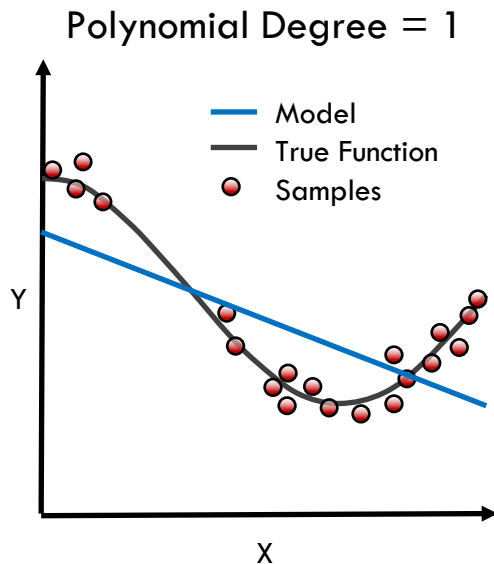
K Value Affects Decision Boundary



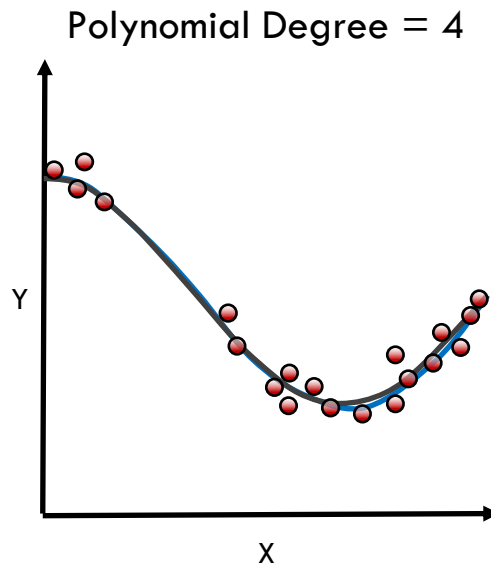
Choosing Between Different Complexities



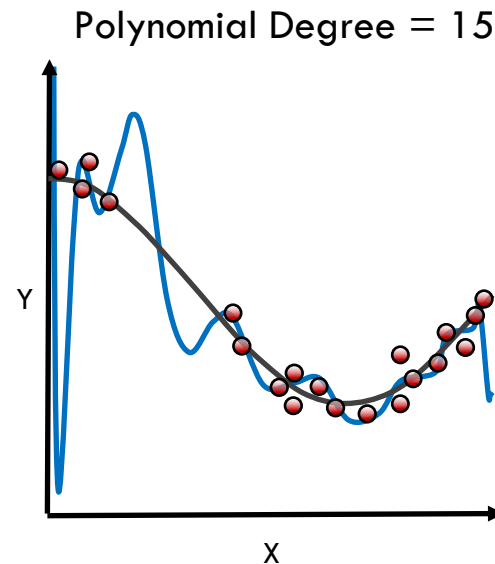
How Well Does the Model Generalize?



Poor at Training
Poor at Predicting

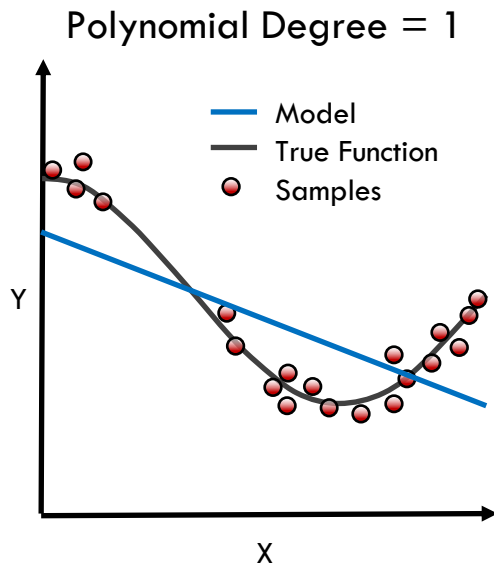


Just Right

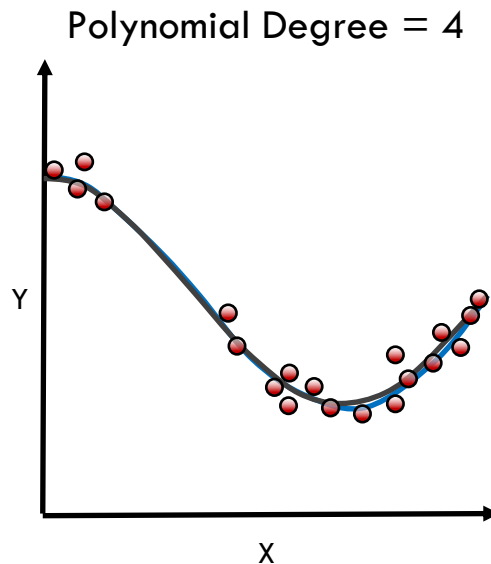


Good at Training
Poor at Predicting

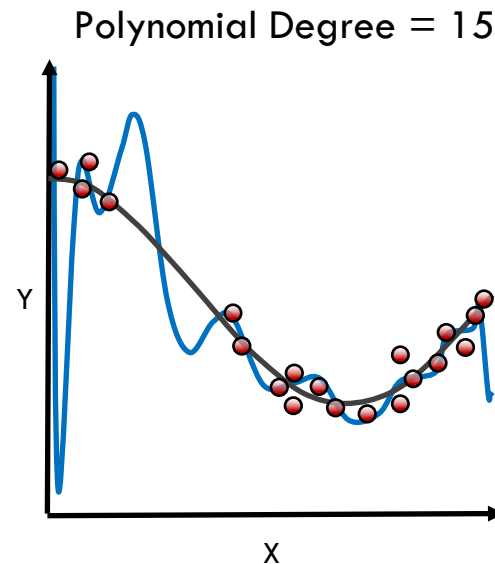
Underfitting vs Overfitting



Underfitting

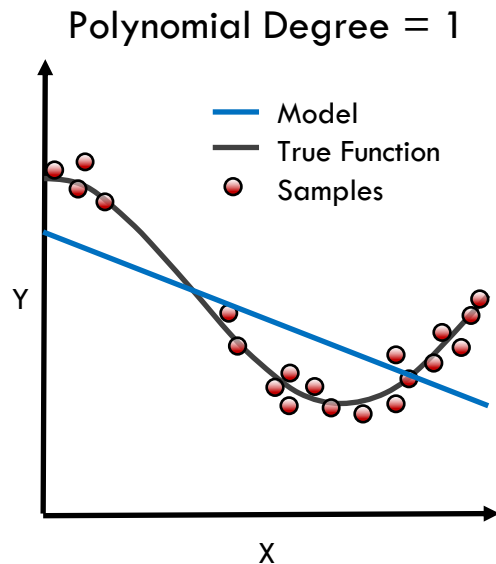


Just Right

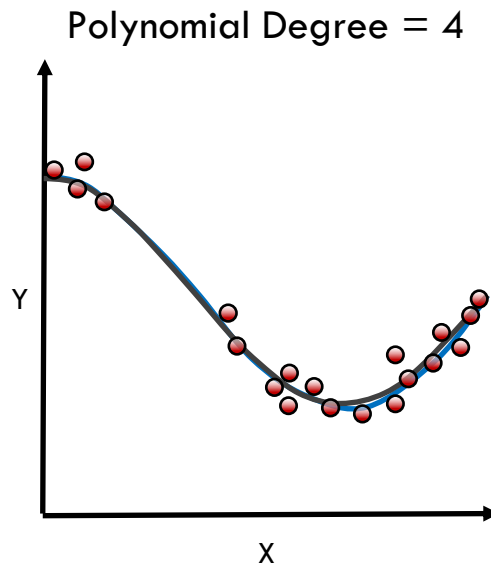


Overfitting

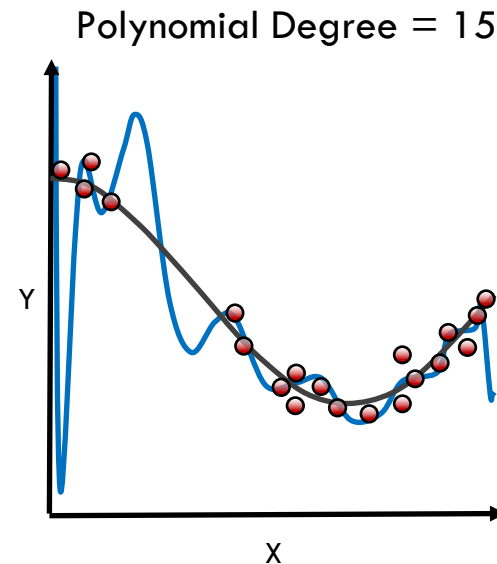
Bias – Variance Tradeoff



**High Bias
Low Variance**



Just Right



**Low Bias
High Variance**

Training and Test Splits

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
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5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
9	2013-03-08	Oz The Great and Powerful	215000000	234911825	Sam Raimi	PG	127
10	2013-05-16	Star Trek Into Darkness	190000000	228778661	J.J. Abrams	PG-13	123
11	2013-11-08	Thor: The Dark World	170000000	206362140	Alan Taylor	PG-13	120
12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

Training and Test Splits

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**Training
Data**

**Test
Data**

Using Training and Test Data

**Training
Data**

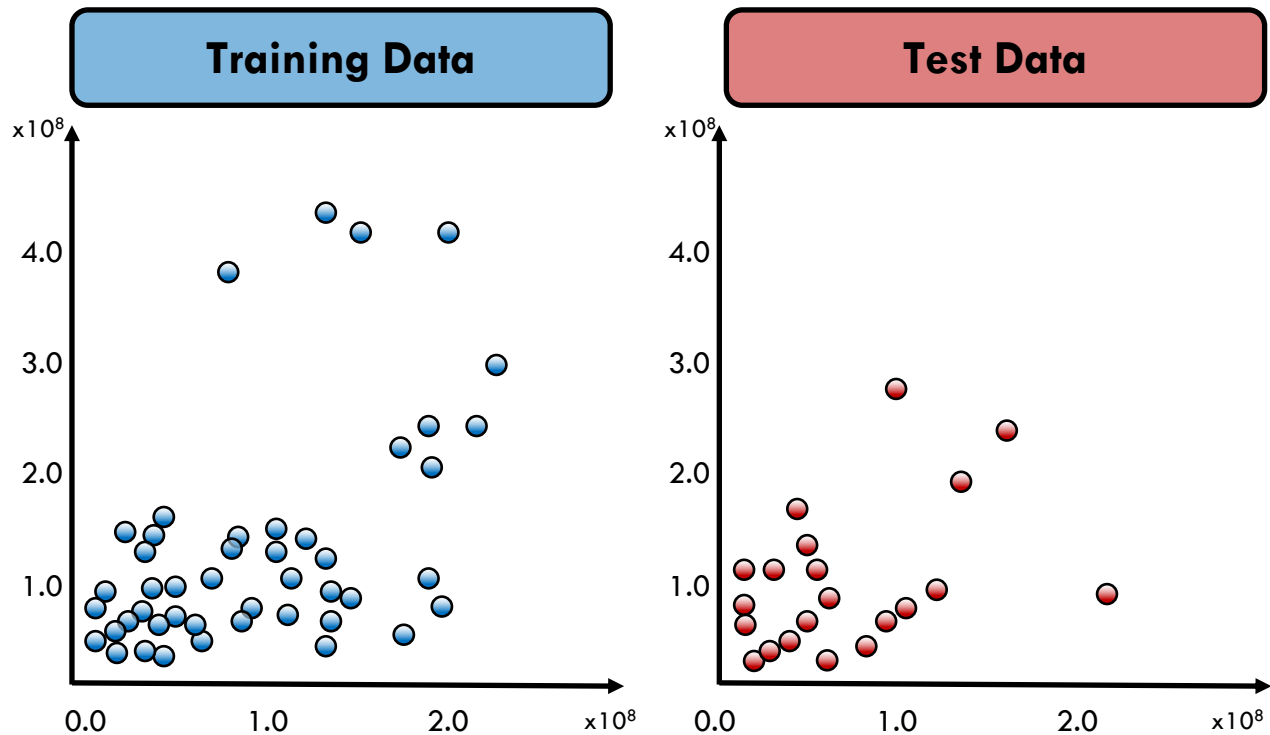
fit the model

**Test
Data**

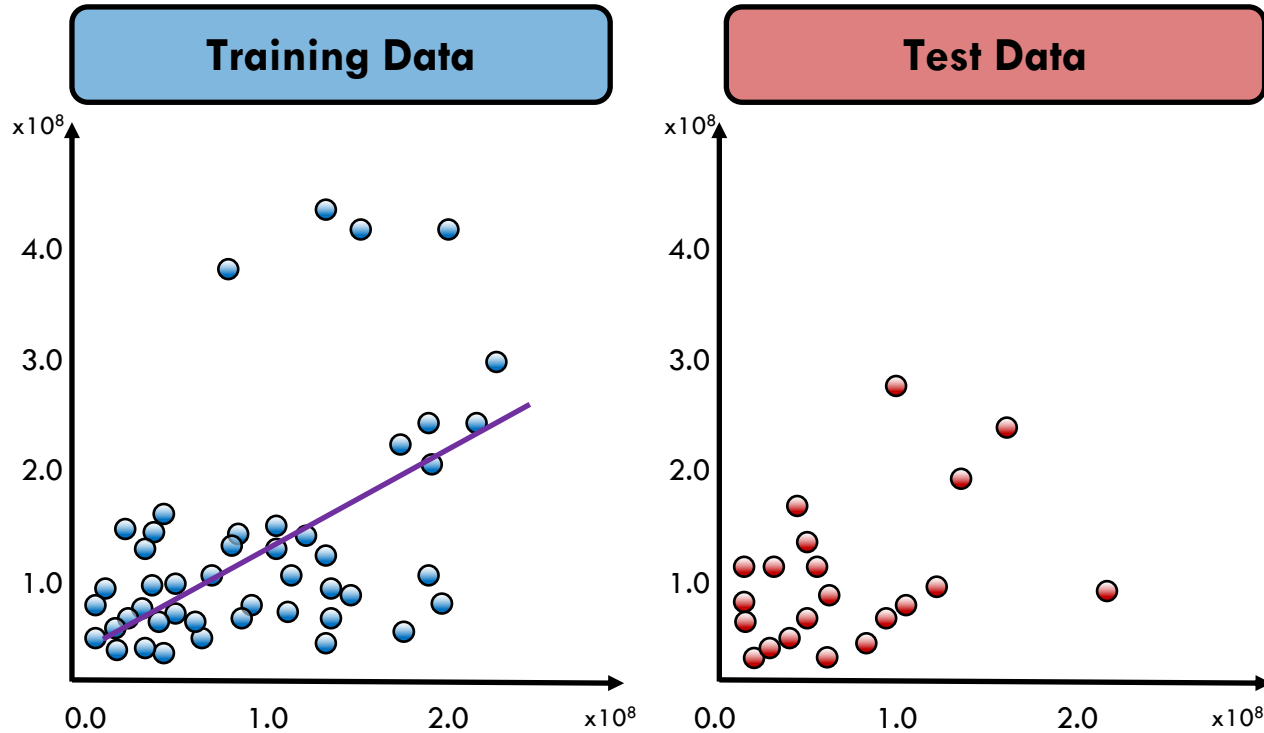
measure performance

- predict label with model
- compare with actual value
- measure error

Using Training and Test Data

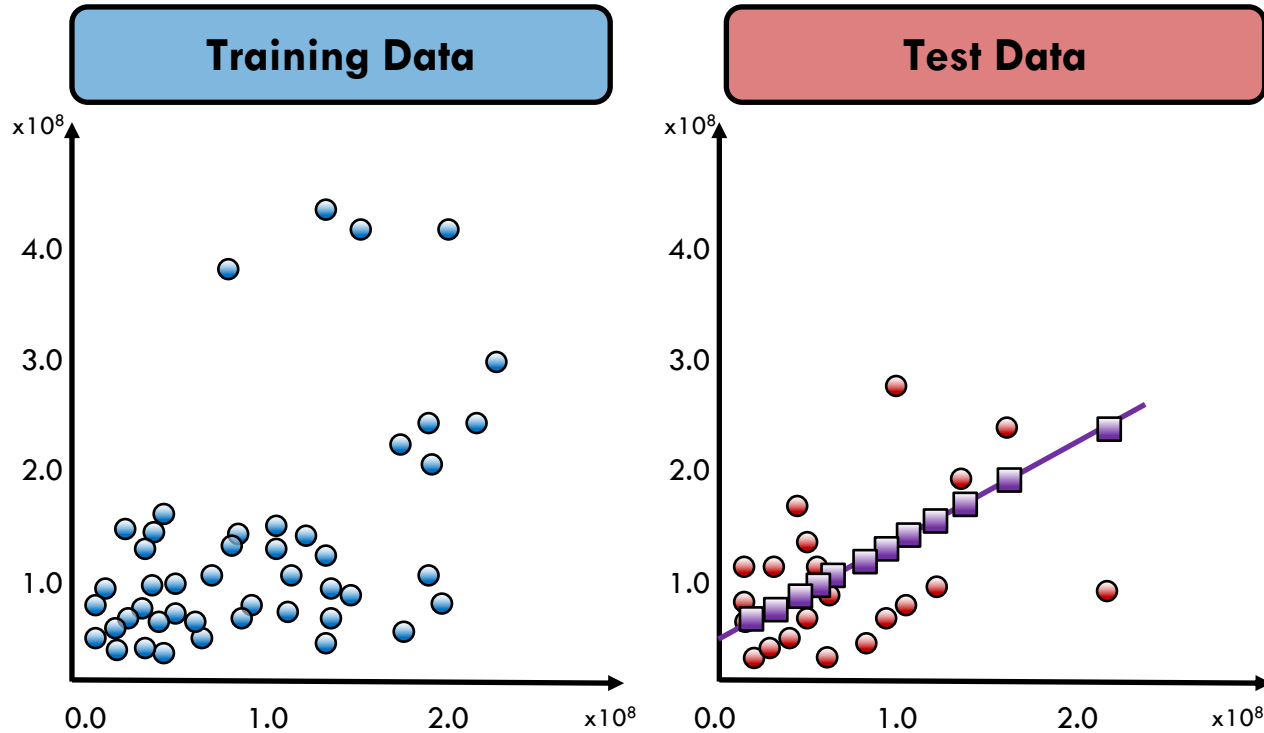


Using Training and Test Data



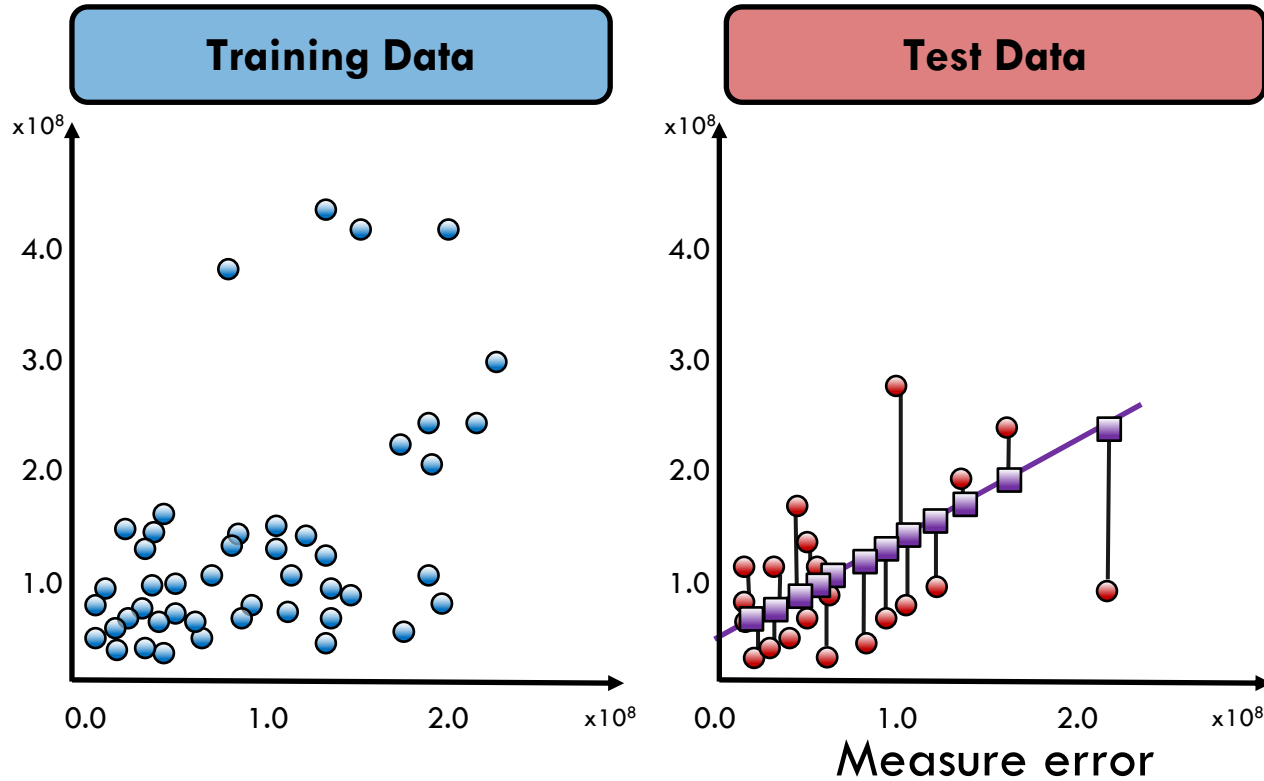
Fit the model

Using Training and Test Data

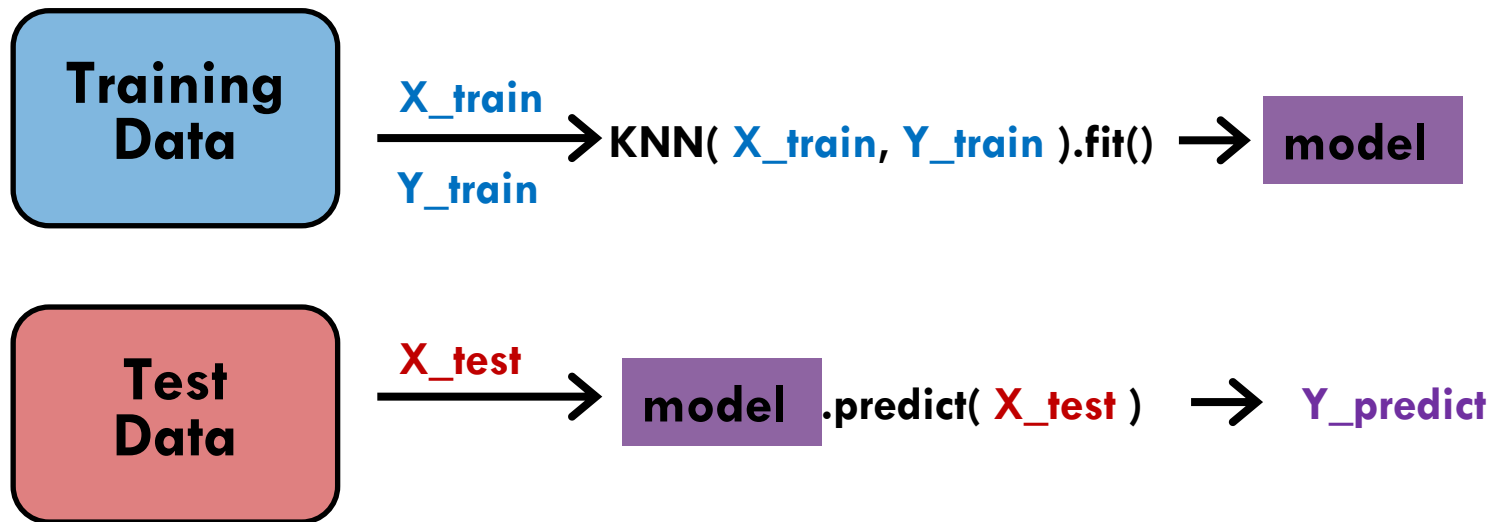


Make predictions

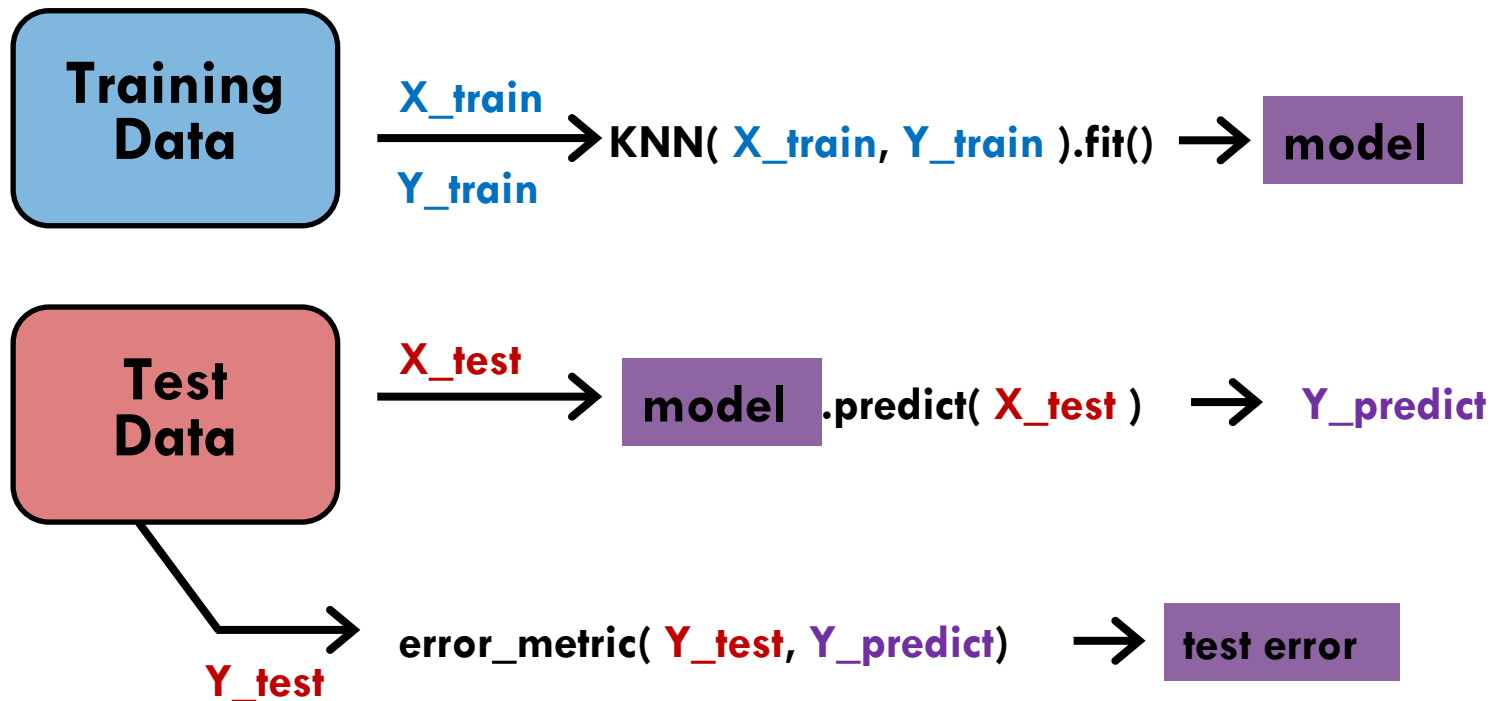
Using Training and Test Data



Fitting Training and Test Data



Fitting Training and Test Data



Train and Test Splitting: The Syntax

Import the train and test split function

```
from sklearn.model_selection import train_test_split
```

Split the data and put 30% into the test set

```
train, test = train_test_split(data, test_size=0.3)
```

Other method for splitting data:

```
from sklearn.model_selection import ShuffleSplit
```

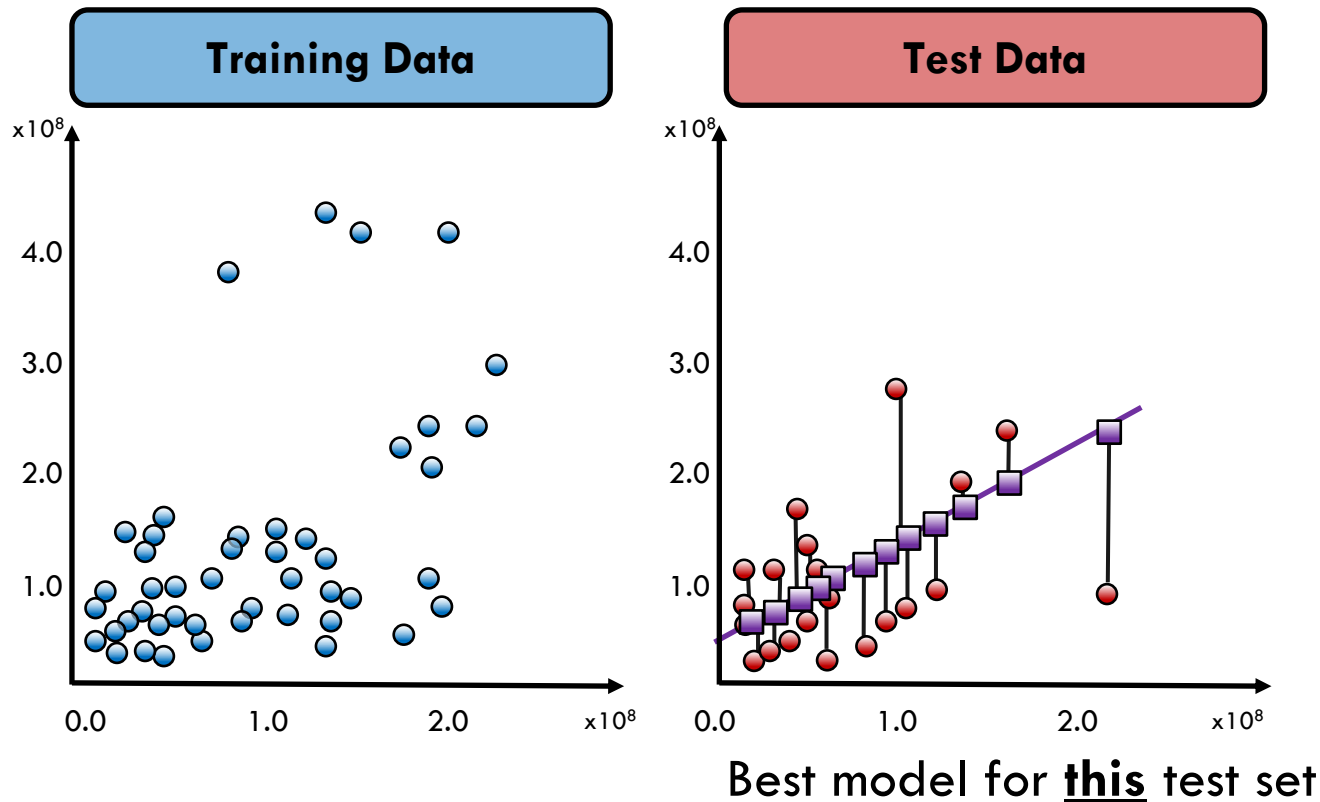
Beyond a Single Test Set: Cross Validation

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8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
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16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

**Training
Data**

**Validation
Data**

Beyond a Single Test Set: Cross Validation



Beyond a Single Test Set: Cross Validation

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3	2013-07-03	Despicable Me 2	76000000	368061265	Pierre CoffinChris Renaud	PG	98
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17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

**Training
Data 1**

**Validation
Data 1**

Beyond a Single Test Set: Cross Validation

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8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
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16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

**Training
Data 2**

**Validation
Data 2**

Beyond a Single Test Set: Cross Validation

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0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
1	2013-05-03	Iron Man 3	200000000	409013994	Shane Black	PG-13	129
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7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
8	2013-05-24	Fast & Furious 6	160000000	238679850	Justin Lin	PG-13	130
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16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

**Validation
Data 3**

**Training
Data 3**

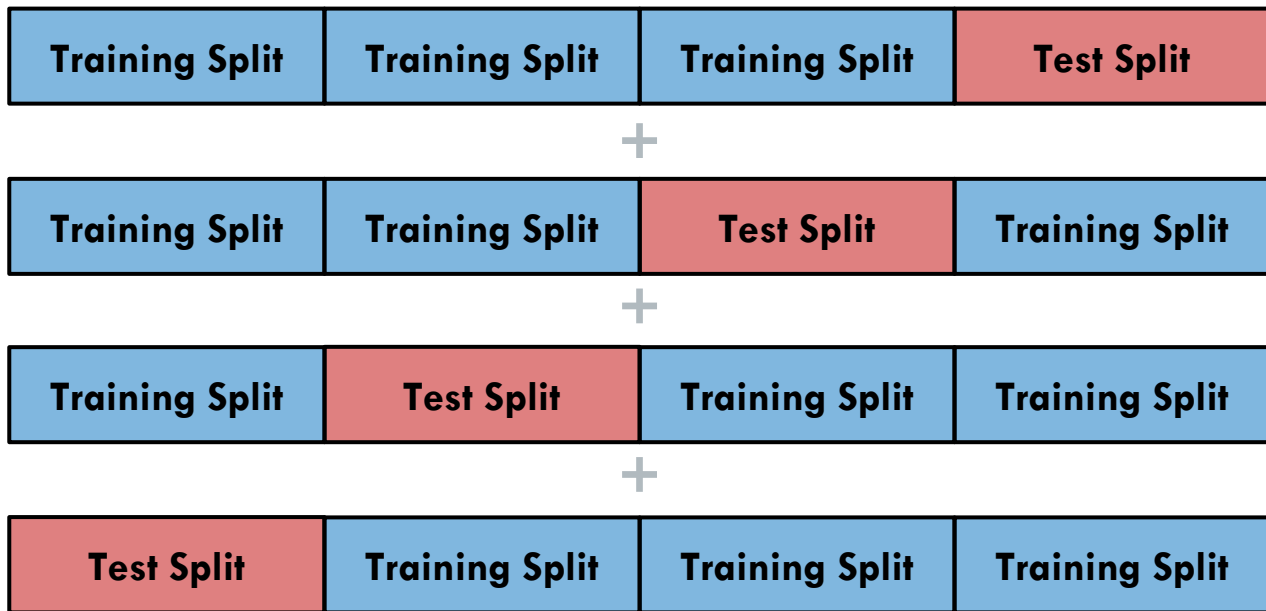
Beyond a Single Test Set: Cross Validation

	Date	Title	Budget	DomesticTotalGross	Director	Rating	Runtime
0	2013-11-22	The Hunger Games: Catching Fire	130000000	424668047	Francis Lawrence	PG-13	146
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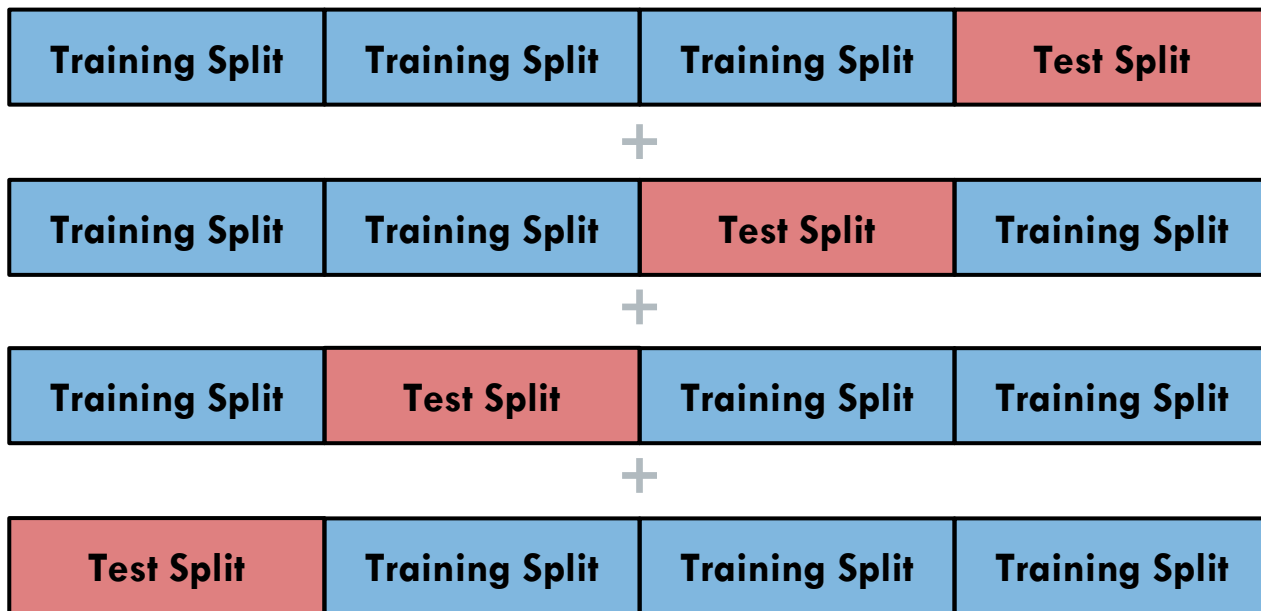
**Validation
Data 4**

**Training
Data 4**

Beyond a Single Test Set: Cross Validation

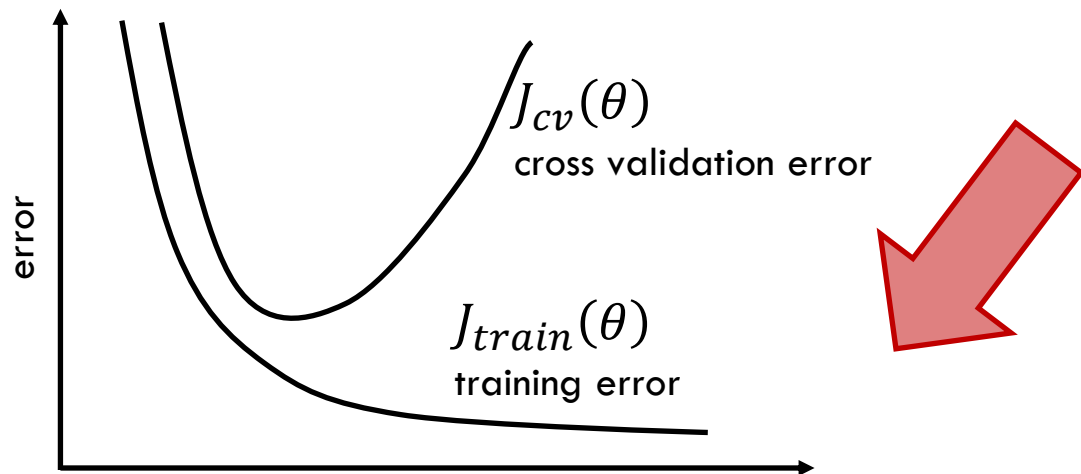


Beyond a Single Test Set: Cross Validation

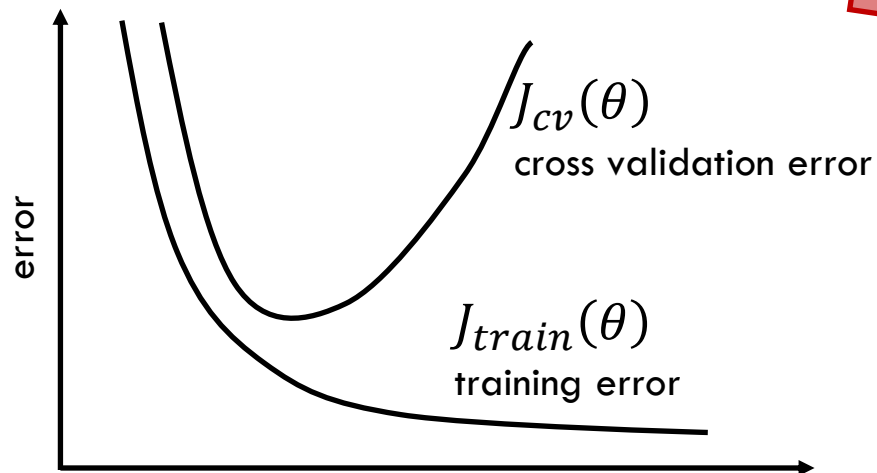


Average cross validation results.

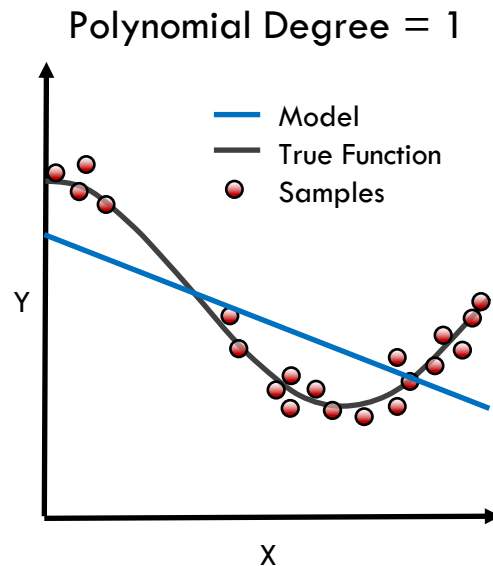
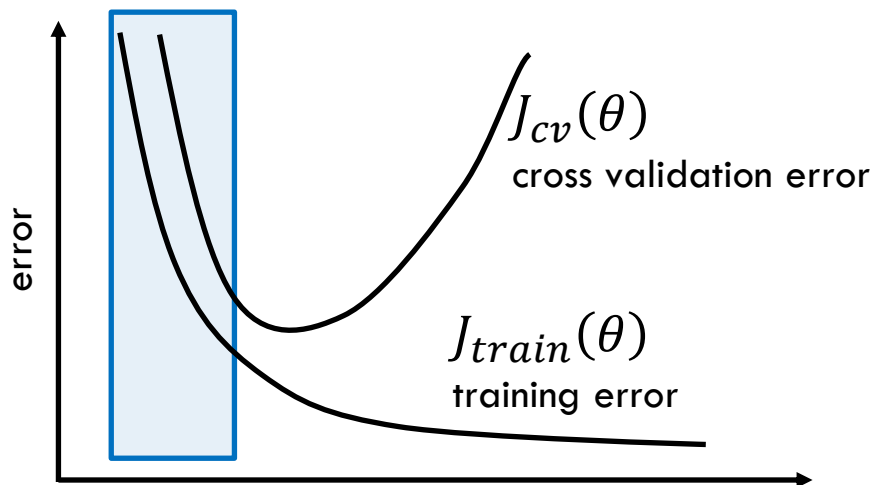
Model Complexity vs Error



Model Complexity vs Error

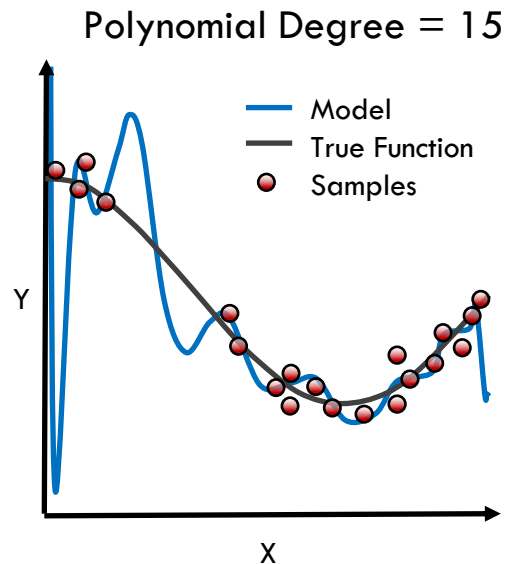
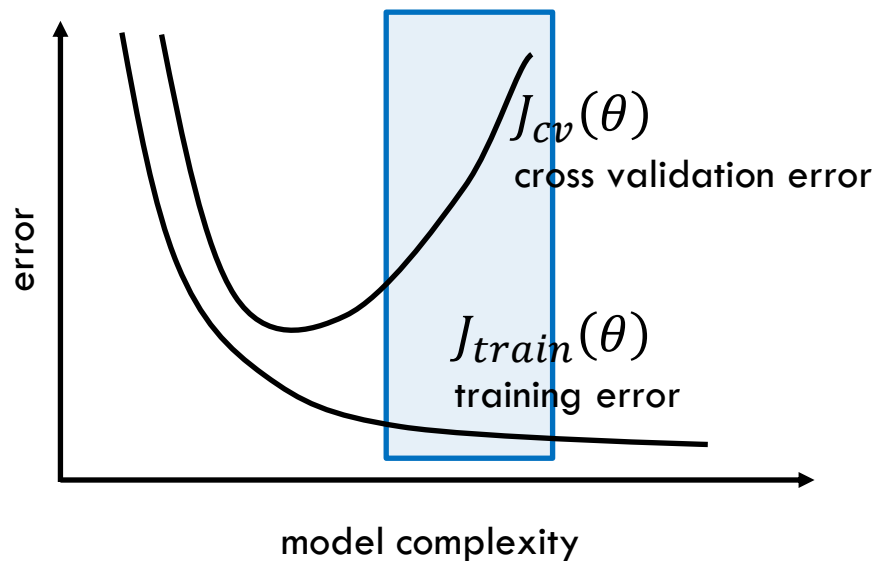


Model Complexity vs Error



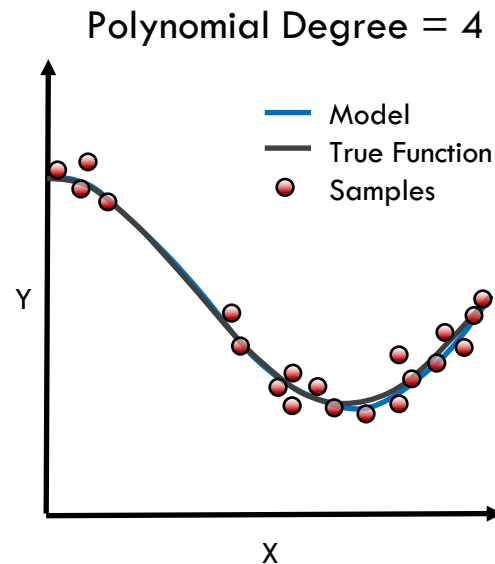
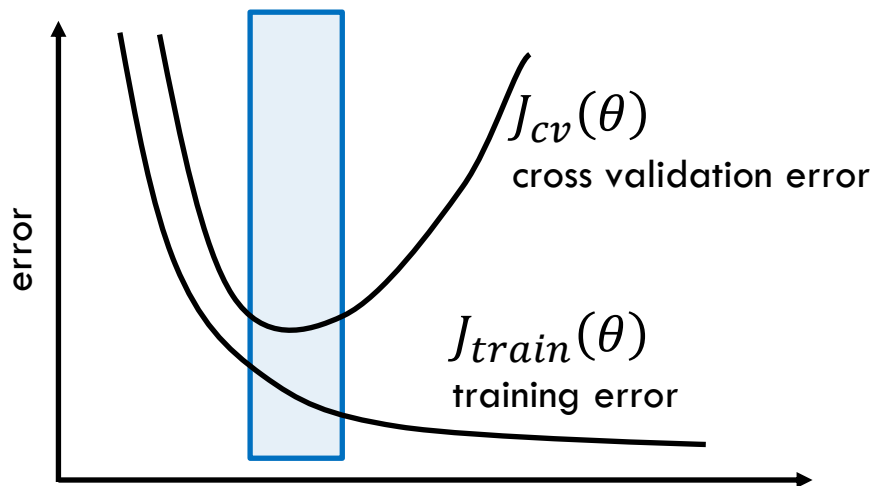
Underfitting: training and cross validation error are high

Model Complexity vs Error



Overfitting: training error is low, cross validation is high

Model Complexity vs Error



Just right: training and cross validation errors are low

Cross Validation: The Syntax

Import the train and test split function

```
from sklearn.model_selection import cross_val_score
```

Perform cross-validation with a given model

```
cross_val = cross_val_score(KNN, X_data, y_data, cv=4,  
                             scoring='neg_mean_squared_error')
```

Other methods for cross validation:

```
from sklearn.model_selection import KFold, StratifiedKFold
```